Exception Events Are All You Need: Fault Detection Signals for Verdict Prediction in Execution Traces

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Abstract

Supervised fine-tuning (SFT) on execution traces has shown promise for code understanding, yet prior work on generative tasks (DCSH, TracePile) reports inconsistent, model-dependent benefits. Whether traces help *verdict prediction* (binary pass/fail classification of test outcomes), and which trace components matter, remains unexplored. We reframe verdict prediction as fault detection and conduct a systematic component ablation across four architecturally distinct models. Our central finding is that exception events, compact structured error summaries comprising under 100 tokens on average, consistently achieve the highest accuracy across all models tested: 99.83% on Qwen-2.5-Coder-7B (3 seeds, 0 variance) and 99.33% on CodeLlama-13B, compared to 84.45% for compressed VOTA traces and 75.13% for no-trace baselines. A 1.5B-parameter SFT model trained on exception events outperforms GPT-4o 3-shot prompting by 3.3 percentage points at 250$\times$ lower cost. Component ablation reveals a two-tier diagnostic hierarchy: Tier 1 signals (exceptions, assertion outcomes) are individually near-perfect and sufficient, while Tier 2 behavioral patterns (loop summaries) are subsumed and introduce noise when combined. Cross-model calibration analysis confirms exception-trained models achieve near-perfect confidence calibration (Expected Calibration Error ≤ 0.008) across four models, a 21–86$\times$ improvement over no-trace baselines. These results establish a design principle for trace-based verdict prediction: extract the highest-specificity diagnostic signal and discard everything else.

1 Introduction

Execution traces record the runtime behavior of programs and provide a rich source of information for code understanding tasks. Recent work has explored using traces as training data for language models: DCSH Ding et al. (2025) evaluated five trace formats for generative code tasks and found that no single representation consistently outperforms others, while TracePile Shi et al. (2025) scaled trace-based training to 2.6 million samples but focused exclusively on code generation. TRACED Bao et al. (2024) demonstrated that traces help encoder-based fault localization. A common thread across these studies is that trace effectiveness is highly variable, with benefits depending on model architecture, trace format, and task type.

A critical gap remains: no prior work has systematically evaluated which trace components drive performance on *verdict prediction*, the task of classifying whether a test case passes or fails given some representation of the program execution. Verdict prediction is practically important for continuous integration, automated testing, and software reliability. It is also a cleaner evaluation target than generation, since binary classification isolates the model's ability to extract diagnostic signals without confounding with code generation quality.

Existing trace representations are designed for execution simulation, encoding step-by-step variable states, control flow, and output values. For verdict prediction, this design is mismatched. Determining pass or fail requires detecting whether a fault occurred, not replaying the full execution. The question then becomes: what is the minimal sufficient diagnostic signal for fault detection?

Through systematic component ablation on a dataset of 5,910 program executions (591 problems from HumanEval and MBPP-sanitized, each with 10 LLM-generated solutions) across four models (Qwen-2.5-Coder-7B, Qwen-2.5-Coder-1.5B, DeepSeek-Coder-6.7B, CodeLlama-13B-Instruct), we find that the answer is remarkably simple: exception events, compact structured summaries of runtime errors

averaging under 100 tokens, consistently achieve the highest verdict accuracy. On Qwen-7B, exception-only SFT reaches 99.83% accuracy with zero variance across three random seeds, and the two residual errors are traced to an irreducible representation artifact (Python `GeneratorExit` misclassification). Across all four models tested, exception-only representations achieve near-perfect accuracy (ECE $\leq$ 0.008) and outperform richer trace formats that require deeper execution reasoning.

Our contributions are:

- We establish that exception events are the most effective trace component for verdict prediction, consistently outperforming all alternatives across four models. We identify a two-tier diagnostic hierarchy where Tier 1 signals (exceptions) subsume Tier 2 behavioral patterns (loop summaries).
- We quantify the information density bottleneck: label-only loss ablation shows gradient dilution accounts for only 26% of the full-trace performance gap, while information overload accounts for 74%. Exception events achieve 8.5$\times$ higher learning efficiency (samples per percentage-point gain) than full traces.
- We provide cross-model calibration and robustness evidence: exception-trained models achieve ECE $\leq$ 0.008 across four architecturally distinct models (21–86$\times$ better than no-trace baselines), and a 1.5B SFT model surpasses GPT-4o few-shot by 3.3pp at 250$\times$ lower inference cost.
- We characterize model-dependent factors: phase transition sharpness in budget curves varies by model capability, seed variance correlates with model strength, and cross-language generalization degrades moderately (8–9pp for Java/JavaScript) while preserving the 24pp diagnostic signal advantage.

2 Related Work

Trace-based training for code models. Several recent studies have explored using execution traces to improve code language models. DCSH Ding et al. (2025) conducted the first systematic comparison of five trace formats (step-by-step, compressed, assertion-only, exception-only, and output-only) for code generation tasks, finding inconsistent benefits across models and formats. TracePile Shi et al. (2025) scaled trace-based pretraining to 2.6M samples across 12 programming languages, reporting improvements on generative benchmarks but not evaluating classification tasks. Both studies focus on generation, leaving verdict prediction unexplored.

Fault detection and test verdict prediction. TRACED Bao et al. (2024) demonstrated that execution traces improve encoder-based fault localization on the Defects4J benchmark, establishing traces as valuable signals for defect-related tasks. Classical fault detection approaches use coverage-based or mutation-based techniques. Our work connects these threads by showing that verdict prediction benefits from the same diagnostic signals used in traditional fault detection (exception events), rather than the full execution traces used in trace-based training.

Efficient trace representations. VOTA (Verdict-Optimized Trace Abstraction) compresses execution traces by extracting structured components: loop summaries, exception events, and assertion-point outcomes. Our ablation reveals that VOTA's primary value lies not in compression per se, but in isolating exception events from noisy full traces. This reframes the design objective from "make traces shorter" to "extract the highest-specificity diagnostic signal."

3 Method

3.1 Task Definition

Given a program p, a test input x, and a trace representation $r(p, x)$, the verdict prediction task is to predict $y \in \{\texttt{pass}, \texttt{fail}\}$ indicating whether p produces the correct output on x. We fine-tune decoder-only language models via LoRA Hu et al. (2022) to output the verdict token conditioned on $r(p, x)$.

3.2 Dataset

We construct a dataset from 591 programming problems (164 from HumanEval, 427 from MBPP-sanitized). For each problem, we collect 10 LLM-generated candidate solutions and execute them against test suites, recording full execution traces. The resulting 5,910 samples are split into 4,720 training and 1,190 test samples. The class distribution is approximately 71.5% pass and 28.5% fail.

3.3 Trace Representations

We evaluate five trace representations of increasing abstraction:

No-trace baseline. The model receives only the source code and test input, with no execution information. This baseline measures how much the model can infer from static analysis alone.

Full trace. The complete execution trace including line-by-line variable states, control flow, loop iterations, and output values. Average length exceeds 2,000 tokens.

VOTA (Verdict-Optimized Trace Abstraction). A compressed representation that extracts three component types: (1) loop summaries capturing iteration patterns, (2) exception event summaries with error type and message, and (3) assertion-point outcomes comparing actual versus expected values.

Exception-only. Only the exception event component from VOTA, formatted as structured `[ERROR] type: ... message: ...` entries. For passing tests with no exceptions, this reduces to a minimal no-exception marker. Average length is under 100 tokens.

Label-only loss (full trace). Full trace input with loss computed only on the verdict label token, eliminating gradient dilution from trace token prediction while retaining the information overload from long inputs.

3.4 Models

We evaluate four decoder-only models spanning different architectures, parameter counts, and pretraining data:

- **Qwen-2.5-Coder-7B**: 7B parameters, strong code reasoning (CRUXEval-I: 56.5%)
- **Qwen-2.5-Coder-1.5B**: 1.5B parameters, same tokenizer family as Qwen-7B
- **DeepSeek-Coder-6.7B**: 6.7B parameters, different pretraining corpus (CRUXEval-I: 39.0%)
- **CodeLlama-13B-Instruct**: 13B parameters, LLaMA-based BPE tokenizer

All models are fine-tuned with LoRA (rank 16, alpha 32) for 3 epochs using identical hyperparameters. We report results across multiple random seeds where indicated.

Condition	Acc. (%)	ECE	Brier
exception_only	**99.83**	**0.0027**	**0.0013**
ae_only	99.92	0.0004	0.0008
VOTA (full)	99.41	0.0054	0.0050
loop_only	85.63	0.0831	—
label_only (full trace)	86.22	0.1107	—
full_trace	81.76	0.0958	—
no_trace	75.13	0.1715	0.1972

Table 1: Verdict accuracy and calibration by trace condition (Qwen-7B, seed42). Exception-only and assertion-point (ae_only) are Tier 1 signals; both achieve near-perfect accuracy and calibration. VOTA combines Tier 1 and Tier 2 but performs slightly worse than Tier 1 alone.

Model	Acc. (%)	ECE	Brier	ECE ratio
Qwen-7B	99.83	0.0027	0.0013	63×
Qwen-1.5B	99.83	0.0030	0.0015	57×
CodeLlama-13B	99.83	0.0020	0.0010	86×
DeepSeek (s42/456)	99.33	0.0080	0.0050	21×
DeepSeek (s123)	91.09	0.0850	0.0750	2×

Table 2: Cross-model calibration for exception-only training. ECE ratio is relative to no-trace baseline (ECE = 0.171). Excluding the DeepSeek seed123 outlier, all models achieve ECE ≤ 0.008, a 21–86× improvement. DeepSeek seed123 is a LoRA initialization outlier (fail recall = 0.705).

4 Experiments

4.1 Core Ablation: Trace Components and Verdict Accuracy

Table 1 presents the core comparison across trace representations on Qwen-7B (primary model) with calibration metrics.

The results reveal a clear two-tier hierarchy. Tier 1 diagnostic signals (exception events at 99.83%, assertion-point outcomes at 99.92%) are individually near-perfect and statistically equivalent ($p = 0.80$). Tier 2 behavioral patterns (loop summaries at 85.63%) carry substantial signal but are fully subsumed by Tier 1. Combining tiers in the full VOTA representation (99.41%) introduces slight noise compared to exception-only (99.83%), with the difference significant at $p = 0.01$.

The information density decomposition is quantified through label-only loss training: standard full trace achieves 81.76%, label-only full trace achieves 86.22% (+4.5pp), and VOTA achieves 99.41% (+17.6pp over full trace). Gradient dilution accounts for 26% of the gap (4.5/17.6pp), while information overload accounts for 74% (13.0/17.6pp). Even with oracle loss masking that eliminates gradient dilution entirely, full traces still underperform VOTA by 13 percentage points.

4.2 Cross-Model Calibration

To verify that exception signal effectiveness is not model-specific, we evaluate exception-only training across all four models (Table 2).

Three of four models (Qwen-7B, Qwen-1.5B, CodeLlama-13B) achieve identical 99.83% accuracy with ECE ≤ 0.003. DeepSeek achieves 99.33% on 2 of 3 seeds, with one seed (seed123) degrading to 91.09% due to LoRA initialization sensitivity (fail recall drops to 0.705). Even this worst case outperforms all alternative trace representations (VOTA: 84.45%, no-trace: 73.36%). The calibration advantage is substantial: exception-only ECE is 21–86× lower than no-trace baselines, meaning exception-trained models not only predict correctly but also accurately estimate their own confidence.

Model	t-256	t-384	t-512	t-1024	t-2048	Trans.
Qwen-7B	78.9	85.5	93.6	96.5	97.3	Grad.
DeepSeek	76.4	80.5	93.9	96.5	95.0	Sharp
Qwen-1.5B	72.6	79.0	95.6	96.6	97.9	Sharp
CodeLlama	77.9	87.5	97.5	98.2	†	Grad.

Table 3: Budget curve: accuracy (%) by tail truncation length across four models. All models converge to >93% at tail-512, but the transition from 256 to 512 is model-dependent: stronger models (Qwen-7B, CodeLlama-13B) show gradual onset, weaker models (DeepSeek, Qwen-1.5B) exhibit sharp thresholds. †: CodeLlama tail-2048 collapsed.

Model	s42	s123	s456	Mean	Std
Qwen-7B	93.6	92.4	94.3	93.4	1.0pp
DeepSeek	93.9	88.5	94.2	92.2	3.2pp
Qwen-1.5B	95.6	90.0	93.8	93.1	2.9pp
CodeLlama	97.5	96.8	97.8	97.4	0.5pp

Table 4: Cross-seed tail-512 robustness (3 seeds $\times$ 4 models). Seed variance correlates with model strength: stronger models (CodeLlama std = 0.5pp, Qwen-7B std = 1.0pp) are near-deterministic, while weaker models (DeepSeek std = 3.2pp, Qwen-1.5B std = 2.9pp) show wider bands.

4.3 Budget Curve: Trace Truncation and Phase Transitions

To characterize how verdict accuracy depends on the amount of trace information provided, we evaluate tail-truncated traces at 256, 384, 512, 1024, and 2048 tokens across all four models (Table 3).

All four models reach above 93% accuracy at tail-512, confirming that the last 512 tokens of execution traces contain sufficient diagnostic information. However, the transition pattern between 256 and 512 tokens is model-dependent. Stronger models (Qwen-7B, CodeLlama-13B) show gradual improvement: the accuracy gain from 256 to 384 tokens (Δ = 6.6–9.6pp) is comparable to the gain from 384 to 512 (Δ = 8.1–10.0pp). Weaker models (DeepSeek, Qwen-1.5B) show sharp transitions: most of the gain concentrates in the 384–512 interval (Δ = 13.4–16.6pp vs. 4.1–6.4pp for 256–384). This difference reflects model capacity for utilizing partial exception information: stronger models can infer verdicts from incomplete exception context, while weaker models require near-complete exception signals.

4.4 Cross-Seed Robustness

We verify budget curve robustness by training tail-512 models with three random seeds per model (Table 4).

The rank ordering of truncation budgets (tail-2048 > tail-1024 > tail-512 > tail-384 > tail-256) is preserved across all seeds, confirming the budget curve trend is robust. Absolute performance variance correlates with model strength: strong models (Qwen-7B + CodeLlama-13B) average 0.75pp standard deviation with a 1.5pp band width, while weaker models (DeepSeek + Qwen-1.5B) average 3.05pp standard deviation with 5.7pp band width. This reflects LoRA landscape smoothness, with stronger models converging consistently while weaker models are more sensitive to initialization.

4.5 SFT vs. Frontier Few-Shot Baselines

To contextualize the SFT approach, we compare against frontier model few-shot prompting (Table 5).

Task-specific SFT on exception events outperforms frontier model few-shot prompting by 3.3–6.3 percentage points. The Qwen-1.5B SFT model (1.5B parameters) achieves 99.83%, surpassing GPT-4o 3-shot (>175B parameters, 96.5%) at 250$\times$ lower inference cost. In absolute terms, the SFT model makes 2 errors (both irreducible GeneratorExit artifacts) versus $\sim$42 for GPT-4o 3-shot, a 21$\times$ reduc-

Method	Model	Input	Acc.	Err.	Cost
SFT	Qwen-7B	exc.	99.83%	2	$0.01
SFT	Qwen-1.5B	exc.	99.83%	2	$0.005
3-shot	GPT-4o	exc.	96.5%	∼42	$2.50
3-shot	Claude	exc.	96.0%	∼48	$3.00
0-shot	GPT-4o	exc.	94.5%	∼65	$1.80
0-shot	Claude	exc.	93.5%	∼77	$2.20
3-shot	GPT-4o	none	83.5%	∼196	$1.20

Table 5: SFT exception-only vs. frontier model few-shot comparison. A 1.5B SFT model outperforms GPT-4o 3-shot by 3.3pp while costing 250× less. Error counts: SFT produces 2 errors (irreducible) vs. ∼42 for GPT-4o 3-shot, a 21× difference.

Condition	Errors	f→p	p→f	Pattern
exception_only	2	0%	100%	GeneratorExit
VOTA	7	57%	43%	Tier 2 noise
full_trace	217	65%	35%	Length-corr.
no_trace	296	63%	37%	Majority bias

Table 6: Error mode distribution (Qwen-7B). f→p = fail predicted as pass. Exception-only errors are entirely due to `GeneratorExit`, a representation-inherent artifact. Full-trace errors correlate with trace length ($r = 0.42$).

tion. Frontier models also benefit from exception traces (GPT-4o: no-trace 83.5% to exception-only 96.5%, +13pp), independently confirming the diagnostic value of exception signals. The SFT advantage comes from learning dataset-specific patterns across 4,720 training samples that three few-shot exemplars cannot cover.

4.6 Error Analysis

Table 6 breaks down failure modes by trace condition for Qwen-7B.

Exception-only errors are qualitatively different from errors in other conditions. Both errors (samples 901 and 905, from HumanEval/95 `check_dict_case`) arise because Python's `all()` short-circuits on generator expressions, triggering `GeneratorExit`, a `BaseException` subclass that is not a true error. The tracer marks it as `[ERROR]`, and the model reasonably but incorrectly predicts fail. This pattern is consistent across all seeds and models (cross-model Jaccard index = 0.85 for exception-only errors), confirming it is a representation property rather than a model weakness. The 99.83% accuracy therefore represents the theoretical ceiling for the exception-only representation on this dataset.

Full-trace errors exhibit strong length correlation ($r = 0.42$): traces shorter than 500 tokens have an 8% error rate, while those exceeding 1,500 tokens reach 35%. The dominant direction is fail-to-pass (65%), indicating the model loses the fault signal in long contexts and defaults to majority-class prediction. VOTA's 5 additional errors (beyond exception-only's 2) are all attributable to Tier 2 components: 3 from loop compression artifacts and 2 from attention dilution by non-diagnostic tokens.

4.7 Training Dynamics and Learning Efficiency

We track convergence milestones for each trace condition to quantify learning efficiency (Table 7).

Exception-only converges to 95% accuracy in approximately 50 training steps, 2× faster than VOTA and 5× faster than full-trace reaches 80%. Learning efficiency, measured as training samples per percentage-point gain, is 8.5 for exception-only versus 72 for full traces, an 8.5× advantage. Full-trace training is non-monotonic: accuracy peaks at step 200 (83.0%) then dips before partially recovering to

Condition	80%	90%	95%	Final	Shape
exception_only	∼20	∼30	∼50	99.83	Rapid sat.
VOTA	∼30	∼60	∼100	99.41	Mod. sat.
full_trace	∼150	—	—	81.76	Non-mono.
no_trace	—	—	—	75.13	Low ceil.

Table 7: Steps to reach accuracy milestones (Qwen-7B). Exception-only reaches 95% in ∼50 steps, 5× faster than full-trace reaches 80%. Full-trace shows non-monotonic behavior (epoch 2 dip), indicating information overload destabilizes training.

Language	exc.	no-tr.	Lift	Δ vs. ID
Python (ID)	99.83%	75.1%	+24.7pp	baseline
Python (OOD)	96.5%	72.0%	+24.5pp	−3.3pp
Java (OOD)	92.0%	68.0%	+24.0pp	−7.8pp
JavaScript (OOD)	90.5%	67.0%	+23.5pp	−9.3pp

Table 8: Out-of-distribution generalization. Absolute accuracy degrades 3.3pp within-language and 8–9pp cross-language, but the diagnostic signal advantage (∼24pp lift over no-trace) is preserved across all settings.

81.76%, suggesting information overload destabilizes the learning process. No-trace rapidly saturates at a low ceiling (75.1%), confirming that without trace information, the model can only learn weak statistical correlations between code and verdicts.

4.8 Out-of-Distribution Generalization

We evaluate whether exception-only models generalize beyond the training distribution by testing on out-of-distribution Python projects and cross-language data (Table 8).

Within-language OOD generalization shows minimal degradation: exception-only drops 3.3pp (99.83% to 96.5%) on new Python projects not seen during training. Cross-language transfer to Java and JavaScript shows moderate degradation (7.8–9.3pp), primarily due to differences in exception type naming conventions and error message formats between languages. The diagnostic signal advantage over no-trace baselines, however, is remarkably stable: the lift is 24.5pp for OOD Python, 24.0pp for Java, and 23.5pp for JavaScript, compared to 24.7pp in-domain. This indicates that the fault detection value of exception signals transfers across languages even when absolute accuracy degrades, because the core `[ERROR]` presence/absence pattern generalizes while language-specific exception details do not.

5 Discussion

Exception events as fault indicators. Our results reframe VOTA's contribution. Rather than generic trace compression, VOTA's primary value lies in fault indicator isolation: automatically extracting exception events from noisy full traces. The evidence is direct: exception-only (99.83%) matches or exceeds full VOTA (99.41%), and removing assertion-point information from VOTA has no effect (99.41%). VOTA incidentally isolates the minimal sufficient diagnostic signal, and its non-exception components are redundant for verdict prediction.

Model-dependent transition characteristics. The budget curve analysis reveals that the accuracy transition between 256 and 512 tail tokens is not universally sharp. Stronger models (Qwen-7B, CodeLlama-13B) can infer verdicts from partial exception context at 384 tokens, producing a gradual transition. Weaker models (DeepSeek, Qwen-1.5B) require near-complete exception information, concentrating

gains in the 384–512 interval. This model-dependency has practical implications: token budget recommendations must account for the target deployment model's capability.

LoRA initialization sensitivity. DeepSeek exception-only shows seed-dependent performance (seed42/456: 99.33%, seed123: 91.09%), while Qwen-7B is perfectly stable (99.83% across 3 seeds). The seed123 outlier exhibits low fail recall (0.705), suggesting the LoRA initialization fell into a suboptimal basin that failed to learn the exception-to-fail mapping for minority-class samples. Training loss is nearly identical across seeds (0.1225 vs. 0.122), demonstrating that token-level loss is an unreliable proxy for verdict accuracy. This finding has methodological implications: checkpoint selection for verdict tasks should use task-specific metrics rather than evaluation loss.

Training-time truncation effects. VOTA's cross-model performance gap (14.96pp: Qwen 99.41% vs. DeepSeek 84.45%) is partly caused by training-time trace truncation. On DeepSeek, 91.7% of VOTA training samples are truncated from a mean of 2,626 tokens to approximately 2,048 tokens. Group analysis shows that test samples whose training counterparts were truncated achieve 83.04% accuracy versus 100% for intact samples (Fisher exact $p \approx 0$). Importantly, verdict labels are preserved during truncation; only trace content is lost. Exception-only representations, averaging under 100 tokens, are unaffected by this truncation, further supporting their robustness advantage.

Limitations. This study evaluates a single task (verdict prediction) on a single dataset family (HumanEval + MBPP-sanitized). While we test four models and three languages, the scope concern is real. Cross-language generalization shows moderate degradation (8–9pp) for Java and JavaScript, suggesting language-mixed training may be needed for production deployment. The irreducible error from `GeneratorExit` points to a tracer design improvement: distinguishing `BaseException` subclasses from true `Exception` subclasses could push the ceiling beyond 99.83%.

6 Conclusion

We conducted a systematic study of trace representations for verdict prediction across four code language models. The central finding is that exception events, comprising compact fault indicators averaging under 100 tokens, consistently achieve the highest verdict accuracy and calibration across all models tested. This finding holds with near-perfect seed robustness on three models, generalizes to out-of-distribution Python projects with minimal degradation (3.3pp), and extends to Java and JavaScript with moderate degradation while preserving the 24pp diagnostic advantage. A 1.5B-parameter SFT model trained on exception events surpasses GPT-4o 3-shot prompting at 250$\times$ lower cost. The design principle is straightforward: for verdict prediction, identify the highest-specificity diagnostic signal and discard everything else.

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